A weighted Cauchy–Schwarz inequality

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Prove that

$$\left(\sum_{j=1}^{n} a_j b_j\right)^2 \leq \left(\sum_{j=1}^{n} j\, a_j^2\right)\left(\sum_{j=1}^{n} \frac{b_j^2}{j}\right)$$

for all real numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$.

Proof. For each j with $1 \leq j \leq n$, we have $j > 0$. Hence $\sqrt{j} > 0$, in particular $\sqrt{j} \neq 0$, and $(\sqrt{j})^2 = j$.

Set

$$x_j = \sqrt{j}\, a_j \qquad \text{and} \qquad y_j = \frac{b_j}{\sqrt{j}}$$

for $j = 1, \ldots, n$. We compute the products and squares of these numbers. First,

$$x_j y_j = (\sqrt{j}\, a_j)\left(\frac{b_j}{\sqrt{j}}\right) = a_j b_j \cdot \frac{\sqrt{j}}{\sqrt{j}} = a_j b_j,$$

since $\sqrt{j} \neq 0$. Summing over j gives

$$\sum_{j=1}^{n} x_j y_j = \sum_{j=1}^{n} a_j b_j.$$

Next,

$$x_j^2 = (\sqrt{j}\, a_j)^2 = (\sqrt{j})^2 a_j^2 = j\, a_j^2,$$

so

$$\sum_{j=1}^{n} x_j^2 = \sum_{j=1}^{n} j\, a_j^2.$$

Also,

$$y_j^2 = \left(\frac{b_j}{\sqrt{j}}\right)^2 = \frac{b_j^2}{(\sqrt{j})^2} = \frac{b_j^2}{j},$$

so

$$\sum_{j=1}^{n} y_j^2 = \sum_{j=1}^{n} \frac{b_j^2}{j}.$$

Now apply the Cauchy–Schwarz inequality to $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$:

$$\left(\sum_{j=1}^{n} x_j y_j\right)^2 \leq \left(\sum_{j=1}^{n} x_j^2\right)\left(\sum_{j=1}^{n} y_j^2\right).$$

Substituting the three identities proved above yields

$$\left(\sum_{j=1}^{n} a_j b_j\right)^2 \leq \left(\sum_{j=1}^{n} j\, a_j^2\right)\left(\sum_{j=1}^{n} \frac{b_j^2}{j}\right),$$

as required. □